MODULE 17 · DOMAIN 4 - PERFORMANCE OPTIMIZATION, QUERYING, AND TRANSFORMATION

Warehouse sizing, scaling and what it costs

Domain 4 begins. Scale up or scale out is the decision, and the exam asks it repeatedly.

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Scale **up** for one slow query. Scale **out** for many queued queries. They fix different problems.

Auto-suspend and auto-resume

- **Auto-suspend** stops a warehouse after a period of inactivity. **Auto-resume** restarts it when a query arrives.
- Every resume bills a **60-second minimum**. A warehouse that resumes twenty times pays that minimum twenty times.
- Suspending **clears the warehouse local disk cache**. The next queries run cold and are slower.
- So the setting is a trade: **short suspend saves idle credits, long suspend preserves the cache**.
- For a warehouse serving frequent short queries, a longer auto-suspend is often **cheaper**, not more expensive.

> ON THE EXAM
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> That last line is counter-intuitive and it was a Domain 1 drill question. Frequent resumes cost more than staying warm.

Scale up or scale out?

SCALE UP

A larger warehouse size

- One query is slow
- More memory, so less spilling to disk
- More local storage for the cache
- Doubles the credits per hour, and often halves the time

SCALE OUT

More clusters

- Many queries are queuing
- More queries run concurrently
- Each query still runs in one cluster
- Does nothing for a single slow query

THE TIEBREAKER

Read the Query Profile. Spilling means scale up. Queuing means scale out. Module 19 reads both off the profile.

Multi-cluster: two modes

- **Maximized mode:** minimum equals maximum. **All clusters run concurrently** from the moment the warehouse resumes.
- **Auto-scale mode:** maximum is greater than minimum. Clusters start and stop between those bounds.
- **The scaling policy does not apply in Maximized mode**, because nothing ever starts or stops.
- Maximized suits steady, predictable high concurrency. Auto-scale suits workloads that fluctuate.
- Multi-cluster warehouses require **Enterprise Edition** or higher.

> ON THE EXAM
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> The mode is decided entirely by whether min equals max. There is no separate setting to choose it.

Standard or Economy scaling policy?

STANDARD - THE DEFAULT

Minimise queuing

- Starts a cluster immediately when a query queues
- Subsequent clusters wait 20 seconds each
- Shuts down after 2 to 3 consecutive checks
- Favours starting clusters over conserving credits

ECONOMY

Conserve credits

- Starts a cluster only if load will keep it busy 6 minutes
- Accepts queuing during that evaluation
- Shuts down after 5 to 6 consecutive checks
- Favours keeping running clusters fully loaded

HOW IT ACTUALLY WORKS

Checks run every minute against the least loaded cluster. With Standard and 5 clusters, starting all five takes at least 80 seconds.

What a multi-cluster warehouse costs

- **Credits equal warehouse size, times running clusters, times time.** A Large with 3 clusters is 24 credits an hour.
- The **60-second minimum applies per cluster**, each time that cluster starts.
- **Maximized mode pays for every cluster all the time** the warehouse is running, whether busy or not.
- Setting the **minimum above 1** keeps that many clusters running permanently. Useful for availability, and never free.

> ON THE EXAM
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> A common accident: setting minimum to 3 for 'reliability' and quietly paying triple for a warehouse that needed one cluster.

Queuing and timeouts

- When a warehouse is at capacity, further queries **queue** rather than failing. Query Profile shows the queued time.
- **STATEMENT_TIMEOUT_IN_SECONDS** aborts a running statement. Default is **172800 seconds, which is 48 hours**.
- **STATEMENT_QUEUED_TIMEOUT_IN_SECONDS** aborts a statement that has waited too long in the queue. **Default is 0, meaning no limit.**
- Both can be set at account, warehouse, user or session level. The **most specific** setting wins.

> ON THE EXAM
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> Those defaults are why a runaway query can burn credits for two days. Setting a sensible STATEMENT_TIMEOUT is basic hygiene.

Generation 2 warehouses

- **Gen2 standard warehouses are generally available**, set with GENERATION = '2'. Snowflake cites roughly 2.1x faster core analytics.
- **Gen2 tops out at 4X-Large.** 5X-Large and 6X-Large are Gen1-only sizes.
- GENERATION applies to **standard warehouses only**, not to Snowpark-optimized ones.
- Converting a running Gen1 warehouse charges for **both** until in-flight queries finish, and it will not auto-suspend meanwhile.
- Behaviour change bundle **2026_03 makes Gen2 the default** for new standard warehouses where available.

> ON THE EXAM
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> The exam is written against the classic sizing model. Learn that as the answer, and know Gen2 exists.

Specialised warehouses

- **Snowpark-optimized warehouses** provide more memory per node. That is what they are for.
- **Standard warehouse memory does not increase with size.** An out-of-memory Snowpark job is not fixed by resizing.
- **Snowpark-optimized memory does not increase above Medium** either. Bigger buys cores, not headroom.
- **Snowflake Adaptive Compute** selects size, cluster count and auto-suspend for you. Newer, and awareness only.

> ON THE EXAM
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> Module 6 built a scenario on the memory rule. It is worth repeating because it contradicts everyone's instinct.

How this module gets tested

THE SYMPTOM	THE ANSWER
One query is slow, and spilling to disk	Scale up. A larger warehouse size
Many queries queuing at peak	Scale out. Multi-cluster, Enterprise Edition
Queuing continues despite multi-cluster	Check the scaling policy. Economy waits 6 minutes
Clusters never start despite the policy	Maximized mode. The policy does not apply
Credits high but the warehouse is mostly idle	Minimum cluster count above 1, or a long auto-suspend
Frequent short queries costing more than expected	The 60-second minimum on every resume

ON THE EXAM

Six symptoms. Notice that three of them are about a setting behaving exactly as documented rather than about a fault.

> A single nightly aggregation over a 12 TB table runs on a Medium warehouse and takes four hours. Query Profile shows 340 GB spilled to local storage and 80 GB spilled to remote storage. There is no queuing.
>
> An engineer proposes converting the warehouse to multi-cluster with a maximum of six clusters.

Will that help, and what should be done?

A Yes. Six clusters give six times the compute, so the query will finish roughly six times faster.

B Yes, but only with the Standard scaling policy. Economy would leave the query queued.

C No. Add a clustering key to the source table, since spilling indicates poor pruning.

D No. A single query runs in one cluster. Scale up instead, which gives more memory and less spilling.

◆ PAUSE AND DECIDE

Will that help, and what should be done?

A Six clusters, six times faster.

The instinct the module exists to correct. A single query cannot be spread across clusters.

B Yes, with the Standard policy.

Right that the policies differ, and no policy makes one query use two clusters.

C Add a clustering key.

Spilling means insufficient memory, not poor pruning. Pruning shows as partitions scanned, which the scenario does not mention.

D One cluster per query. Scale up.

Spilling to local and remote storage is the classic scale-up signal. A larger size gives more memory and more local disk.

THE SIGNAL

Remote spilling especially is a memory shortage. It is the single clearest scale-up signal in the Query Profile.

> A BI warehouse is multi-cluster with a minimum of 1 and a maximum of 8, using the Economy scaling policy. Between 9 and 10 each morning, dashboards are slow and Query Profile shows queries queuing for two to four minutes.
>
> Monitoring shows the warehouse rarely runs more than two clusters, even at the busiest point.

What is the most likely cause?

A. The maximum of 8 is too low for the load. Raise it to 16.

B. The warehouse is in Maximized mode, so it cannot scale beyond its minimum.

C. Economy starts a cluster only if load will keep it busy 6 minutes. A one-hour peak of short queries rarely qualifies.

D. Queuing is unrelated to clusters. Increase STATEMENT_QUEUED_TIMEOUT_IN_SECONDS.

◆ PAUSE AND DECIDE

What is the most likely cause?

A Raise the maximum to 16.

It is only using two of the eight it already has. More headroom it will not use changes nothing.

B It is in Maximized mode.

Minimum 1 and maximum 8 are different, so it is auto-scale. Maximized needs them equal.

C Economy needs 6 minutes of sustained load.

Documented. Short dashboard queries rarely justify a cluster under that rule, so queuing is tolerated by design.

D Raise the queued timeout.

That aborts queries that wait too long. It makes the symptom worse rather than fixing the cause.

THE SIGNAL

'Rarely more than two clusters, with 8 available' is the tell. The ceiling is not the constraint; the policy is.

> A team creates a multi-cluster warehouse with MIN_CLUSTER_COUNT = 4 and MAX_CLUSTER_COUNT = 4, and sets SCALING_POLICY = ECONOMY expecting to save credits during quiet periods.
>
> The monthly bill is four times what they expected, and monitoring shows four clusters running whenever the warehouse is up, including overnight when almost nothing runs.

What is happening?

A. Economy is misconfigured. It must be set before the clusters are created to take effect.

B. Minimum equals maximum, so this is Maximized mode. All clusters always run and the scaling policy does not apply.

C. Economy only reduces clusters after 5 to 6 checks, so overnight the checks never complete.

D. Four clusters are the minimum billable unit above X-Small. Reduce the warehouse size instead.

◆ PAUSE AND DECIDE

What is happening?

A Economy set too late.

Scaling policy can be altered at any time. Ordering is not the issue.

B Min equals max, so Maximized. The policy does not apply.

The mode is decided by the counts. With them equal, all clusters run concurrently and nothing ever starts or stops.

C Overnight checks never complete.

Plausible-sounding, and the checks are not running at all. In Maximized mode there is no scale-down process.

D Four clusters is a minimum billable unit.

Invented. A single-cluster warehouse of any size is perfectly valid.

THE SIGNAL

There is no setting that selects the mode. Equal counts mean Maximized, and the scaling policy is then silently ignored.

> A team runs a very large nightly transformation on a 5X-Large standard warehouse. Having read about the performance improvement, they attempt to convert it to Generation 2 by setting GENERATION = '2'.
>
> The statement fails. They ask whether Gen2 is unavailable in their region.

What is the most likely explanation?

A Gen2 tops out at 4X-Large. 5X-Large and 6X-Large are Gen1-only sizes.

B Gen2 requires Business Critical Edition, which the account does not have.

C GENERATION can only be set at creation. Recreate the warehouse with the clause included.

D Gen2 requires the warehouse to be suspended, and theirs was running.

◆ PAUSE AND DECIDE

What is the most likely explanation?

A Gen2 tops out at 4X-Large.

Documented. 5X-Large and 6X-Large exist only as Gen1 sizes, so a 5X-Large cannot be converted.

B Gen2 needs Business Critical.

It is not edition-gated. Warehouse generation is unrelated to the edition matrix.

C GENERATION is creation-only.

An existing warehouse can be converted, though converting a running one charges for both until in-flight queries finish.

D It must be suspended first.

Conversion works on a running warehouse. The documented consequence is double billing, not a failure.

THE SIGNAL

To use Gen2 they must drop to 4X-Large. Whether that is a net gain depends on whether the 2.1x claim offsets the smaller size.

What to carry into the next module

- **Scale up for one slow query. Scale out for many queued queries.** A single query runs in one cluster.
- **Min equals max is Maximized**, and the scaling policy is then ignored. Max above min is auto-scale.
- **Standard starts immediately, Economy waits 6 minutes.** Shutdown after 2 to 3 checks, or 5 to 6.
- **Credits equal size times clusters times time.** The 60-second minimum applies per cluster.
- **Gen2 stops at 4X-Large.** Standard warehouse memory does not grow with size at all.

> ON THE EXAM
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> Next: the three caches, and a decision tree for why a query is slow.